MIL-S-11510E
2 August 1978
SUPERSEDING
MIL-S-11510D
1 September 1970

MILITARY SPECIFICATION

SCALE, BEAM INDICATING: WEIGHING, SPECIFIC GRAVITY; CAPACITY 0.01 TO NOT LESS THAN 200 GRAMS, THREE OR FOUR WEIGHT BEAMS; 0.01 GRAM SENSITIVITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a three- or four-beam type scale with an adjustable platform.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
AAA-S-121	- Scale; Weighing, General Specifications for.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-T-60	- Tape: Packaging, Waterproof.

FSC 6670

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking for US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The scale shall have a capacity of 0.01 gram to not less than 200 grams, and a sensitivity of 0.01 gram at the capacity load. The scale shall be in accordance with AAA-S-121, shall be as shown in Figures 1 or 2 (see 6.3), and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.3 Weight beam. The weight beam shall be equipped as follows:

(a) Two or three bars with hanging, sliding poises, or riders.
(b) Fractional bar having a capacity of 1 gram, graduated in 0.01-gram increments, and provided with a sliding poise or rider.
(c) An adjustment for setting the zero point.
(d) An adjustment means for leveling the scale when out of level condition affects accuracy.

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3.4 Suspension assembly.

3.4.1 Stirrup. The stirrup shall be corrosion-resisting steel approximately 8-inches high and approximately 4-inches wide.

3.4.2 Pan. The pan shall be made from aluminum, polypropylene, or corrosion-resisting steel and shall be removable from the stirrup, or movable to a vertical position within the stirrup.

3.4.3 Platform. The platform for specific gravity determinations shall be adjustable.

3.5 Plastic cover. When specified (see 6.2), a plastic cover shall be provided.

3.6 Damping. Unless otherwise specified (see 6.2), damping shall be incorporated into the scale. The following methods are acceptable: Magnetic, air, or oil.

3.7 Identification marking. The scale shall be identified in accordance with MIL-STD-130.

3.8 Treatment and painting. Unless otherwise specified (see 6.2), treatment and painting shall be in accordance with the contractor's standard practice.

3.9 Workmanship. All parts, components, and assemblies of the scale including castings, forgings, molded parts, stampings, and machined surfaces shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous materials. External surfaces shall be free from burrs, sharp edges, and corners, except when sharp edges and corners are required.

3.9.1 Castings and forgings. Rework of castings and forgings shall be limited to procedures which do not reduce mechanical properties or effect function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Samples. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 1.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The scale shall be examined as specified herein for the following defects:

101. Characteristics not as specified.
102. Material not as specified.
103. Components missing, incomplete, or not as specified.
104. Identification marking missing, incomplete, or illegible.
105. Workmanship not as specified.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

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4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106.	Materials or containers not as specified. Each incorrect material or container shall constitute one defect.
107.	Unprotected metal surfaces not coated with preservative and wrapped or covered with barrier material as specified.
108.	Components not cushioned as specified.
109.	Void spaces in box not filled with cushioning material as specified.
110.	Fiber box closure not Method V as specified.
111.	Strapping not as specified.
112.	Marking missing, illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The scales shall be disassembled the minimum necessary to reduce cube and to safeguard parts vulnerable to damage or loss in accordance with the recommendations of the contractor.

5.1.1.2 Unprotected surfaces. Unprotected metal surfaces of the scale requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-7 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116. Coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, Type II, Grade A, Class 2, secured with tape conforming to PPP-T-60, Type IV.

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5.1.1.3 Scale. The components comprising a complete scale shall be wrapped with cushioning material conforming to PPP-C-843, Type II, Class A, and preserved together in a close-fitting box conforming to PPP-B-636, W6c. Void spaces in the box shall be filled with the cushioning specified herein, to prevent movement and damage to the contents. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.2 Commercial. Each complete scale shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. The scales, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2) strapping shall be Finish B. When specified (see 6.2) strapping shall be Finish A.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for Level A except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. Scales preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The scale is intended for making laboratory-type weight determinations of quantities (0.01 to not less than 200 grams) to a sensitivity of 0.01 gram and for specific gravity determinations.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a plastic cover is required (see 3.5).
(c) When magnetic damping is not required (see 3.6).
(d) When treatment and painting shall be other than as specified (see 3.8).
(e) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(f) When Finish A strapping is required (see 5.2.1).

6.3 Information figures. Figures 1 and 2 show types of scales which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other scales which conform to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

Custodians:
Army - ME
Air Force - 99

Review activities:
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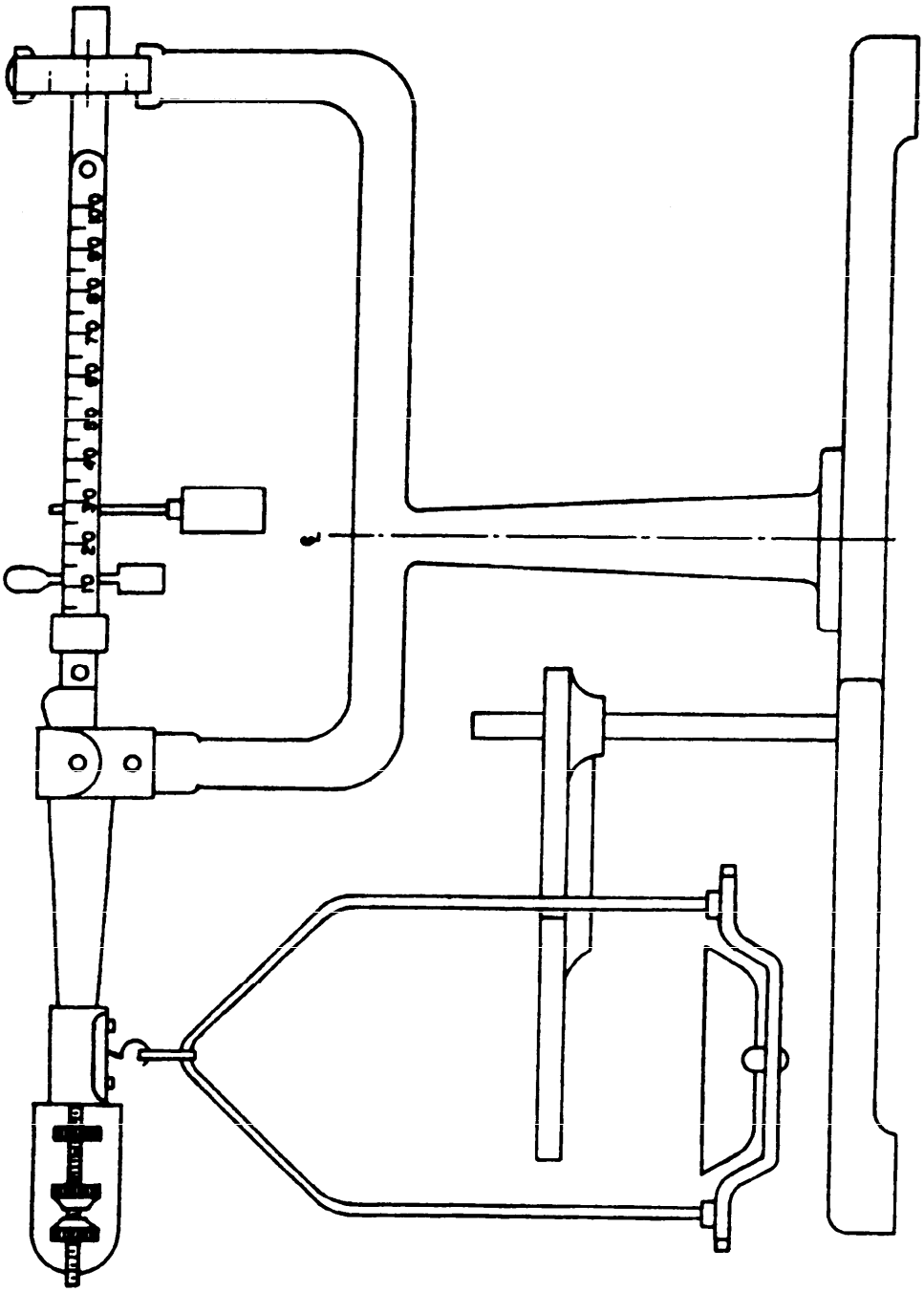
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X-1009A

FIGURE 1. Scale, beam indicating.

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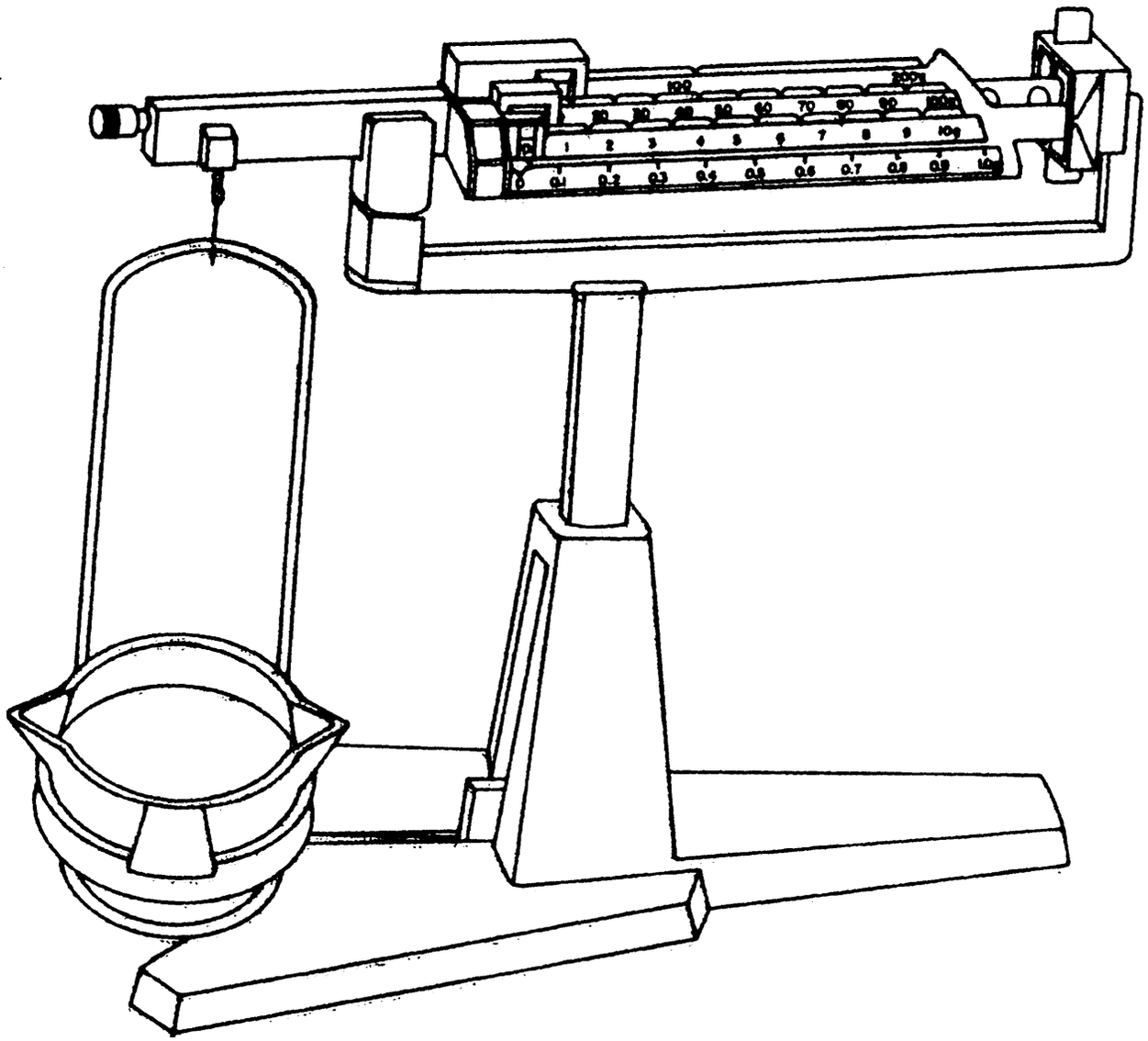


FIGURE 2. Scale, beam indicating.

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DOCUMENT IDENTIFIER AND TITLE MIL-S-11510E Scale, Beam Indicating: Weighing, Specific Gravity; Cap. 0.01 to Not Less Than 200 Grams, etc.

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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